

Zhong Guo

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PROFESSIONAL SKILLS

Summary: Highly skilled and meticulous Researcher with 8 years of experience in Reinforcement Learning/Machine Learning, System Optimal Control, Data Analysis, and Numerical Optimization applied to Building HVAC Systems, Smart Grid, and Virtual Energy Storage.

Software/ Programming Languages /OS: Python, PyTorch, and TensorFlow; EnergyPlus and Modelica, MATLAB and Simulink; Microsoft Office (Excel) and LaTeX; Windows and Linux; PostgreSQL, GitHub and GitLab.

Language Proficiency: English (full professional proficiency); Chinese-Mandarin (native speaker).

EDUCATION BACKGROUND

Doctor of Philosophy, Mechanical Engineering (GPA: 3.79/4.0) **Jan. 2019 — Dec. 2023**

University of Florida, Gainesville, FL.

- PhD Thesis: Reinforcement Learning for Optimal Control of District Cooling Energy Plant.
- Advisor: Prabir Barooah. Committee members: Sean Meyn, Warren Dixon, Yu Wang, Herbert Ingley

Master of Science, Mechanical Engineering (GPA: 3.80/4.0) **Aug. 2017 — Jan. 2019**

University of Florida, Gainesville, FL.

- Master Focus: Dynamics, Systems, and Control.

Bachelor of Science, Machinery Design Manufacturing and Automation (GPA: 3.70/4.0)

Henan Univ. of Sci. & Tech., China.

Sept. 2013 — Jul. 2017

- Focus: Mechanical Design and Automation.

RESEARCH & TEACHING EXPERIENCES

Senior Building Systems Engineer:

Apr. 2024 – present

- Maintain, update, and modernize Trace 3D Plus.
- Trace engine acceleration via High Performance Computing (HPC).
- Optimal control of Water Source Heat Map using MPC.
- Liquid desiccant Dedicated Outdoor Air System (DOAS) modeling and evaluation.

Graduate Researcher:

Jan. 2019 – Mar. 2024

University of Florida, Department of Mechanical and Aerospace Engineering, DiCE lab, Gainesville, FL.

- Finished 4 projects, published 9 papers, and delivered 3 presentations at international conferences.
- Invited to review 11 journal papers and 9 conference papers. Built and maintained the lab database.
- Mentored 1 Master student, leading him to admission to a Ph.D. program.

Project 1: Optimal control of HVAC Systems in commercial buildings. The power consumption of HVAC systems was reduced by over 10% and indoor climate quality was improved by implementing hierarchical Model Predictive Control (MPC) to take advantage of the characteristics of HVAC systems and building thermal dynamics. Both the building thermal dynamics modeling paper and the controlling paper were recognized as h-index papers. In-depth understanding of building HVAC systems and heating & cooling technologies was developed, and hands-on control experiences on large BAS (Building Automation Systems) were acquired. I was also the database administrator to collect, clean, and analyze building data for thermal model identification and control algorithm design.

Project 2: Characterize the energy storage capacity of commercial buildings using PSD (Power Spectral Density). Heavy quantitative analysis was performed and a new probabilistic perspective of quantifying the potential of demand response from commercial buildings was formulated to counteract the penetration of renewable energy in the grid. The appealing techno-economic effects were analyzed.

Project 3: Optimal control of chiller plants using RL (Reinforcement Learning). The electricity cost of the chiller plant was reduced by 10% through optimizing its control using reinforcement learning. The building and chiller plant were modeled based on TRNSYS, EnergyPlus, and Modelica Buildings Library with the applications of data cleaning/analysis and model identification. The designed chiller plant model's energy efficiency is in accordance with ASHRAE Standard 90.1. A modified Q-learning-based algorithm was proposed, and the RL controller was trained using PyTorch and TensorFlow.

Project 4: Improve building indoor climate comfort using PCN (Personal Comfort Node). Developed hands-on hardware engineering experiences: assembling Raspberry Pi, Arduino, and XBee; soldering on circuit boards; processing signals; documenting systematically.

Teaching Assistant:

Aug. 2019 – Aug. 2020

University of Florida, Department of Mechanical and Aerospace Engineering, Gainesville, FL.

- Courses: Mechanical Engineering Control; Vibrations; Refrigeration and Air Conditioning Fundamentals.
- Assisted professors, prepared lecture slides, held office hours, and evaluated homework and exams.

Research Assistant:

Jan. 2018 – Dec. 2018

University of Florida, Department of Mechanical and Aerospace Engineering, DiCE lab, Gainesville, FL

- Was a Master student to help PhD students with lab projects. Work is highly praised by lab mates and academic advisors. Admitted into the PhD program at the same lab.

Undergraduate Researcher Assistant:

May 2014 – May 2016

Gear Manufacture and Equipment Center, Henan Univ. of Sci. & Tech, Luoyang, Henan, China

- Worked as team leader in the Student Research Training Program project of “Design of Portable Meshing Tester for High-Speed Small Module Spiral Bevel Gear”. Concluded the project with a journal paper and a patent. Our team was awarded the first prize.

PROFESSIONAL EXPERIENCES

Structural Design Engineer Intern

Jun. 2016 – Aug. 2016

Technology Center of Zhengzhou Yutong Bus Co, Ltd, Zhengzhou, Henan, China

- Designed and optimized bus components using CATIA, selected as presenter of our group work in company meetings. Our design was adopted for the company's next bus model. Participated in and facilitated 4 workshops in the company.

Outfitting Engineer Intern

Dec. 2015 – Feb. 2016

Zhengzhou Yutong Heavy Industries Co, Ltd, Zhengzhou, Henan, China

- Designed and optimized vehicle outfit which reduced the cost by 5%. Praised by the team leader and received an intern invitation for the subsequent summer.

PUBLICATIONS

Journals

1. Austin R. Coffman, **Zhong Guo**, and Prabir Barooah, "Characterizing capacity of flexible loads for providing grid support", *IEEE Transactions on Power Systems* 36 (3), 2428-2437, Oct. 2020.
2. **Zhong Guo**, Austin R. Coffman, Jeffrey Munk, Piljae Im, and Prabir Barooah, "Aggregation and data-driven identification of building thermal dynamic model and unmeasured disturbance", *Energy and Buildings* 231, Jan. 2021.
3. Naren Srivaths Raman, Rahul Umashankar Chaturvedi, **Zhong Guo**, and Prabir Barooah, "Model predictive control-based hierarchical control of a multi-zone commercial HVAC system", *Journal of Engineering for Sustainable Buildings and Cities* 2 (2), 021005, May. 2021.
4. Joseph Rosenberger, **Zhong Guo**, Austin Coffman, Duzgun Agdas, and Prabir Barooah, "An Open-Source Platform for Indoor Environment Monitoring with Participatory Comfort Sensing", *Sensors* 23 (1), 364, Dec. 2022.
5. **Zhong Guo**, Aditya Chaudhari, Austin R. Coffman, and Prabir Barooah, "Optimal Control of District Cooling Energy Plant with Reinforcement Learning and MPC", *ASME Journal of Engineering for Sustainable Buildings and Cities* (1-16), Nov. 2023.
6. Saptarshi Bhattacharya, Syed Ahsan Raza Naqvi, Mingzhe Liu, Veronica Adetola, Scott Munns, Kaustubh Phalak, Zheng O'Neill, Caleb J. Calfa, **Zhong Guo**, "On predictive device-level controls for water source heat pumps in multi-family residential buildings", *Energy and Buildings*, Volume 351, 2026.

Conferences (also the presenter)

1. **Zhong Guo**, Austin R. Coffman, Jeffrey Munk, Piljae Im and Prabir Barooah, "Identification of aggregate building thermal dynamic model and unmeasured internal heat load from data", *58th IEEE Conference on Decision and Control*, Nice, France, Dec. 2019.
2. Austin R. Coffman, **Zhong Guo**, and Prabir Barooah, "Capacity of flexible loads for grid support: statistical characterization for long term planning", *2020 American Control Conference*, 533-538, Denver, CO, USA, Jul. 2020.
3. **Zhong Guo**, Austin R. Coffman, and Prabir Barooah, "Reinforcement learning for optimal control of a District Cooling Energy Plant", *2022 American Control Conference*, 533-538, Atlanta, GA, USA, Jun. 2022.

SELECTED HONORS & AWARDS

Outstanding Service as Captain of Chinese Soccer Team at University of Florida	Dec. 2023
Achievement Awards Scholarship for New Engineering Graduate Students	Aug. 2017
Outstanding Graduate in Henan Province (province-wide award)	Feb. 2017
Outstanding Graduate in Henan University of Science & Technology.	Nov. 2016
National Endeavor Scholarship (nation-wide award)	Oct. 2016
Excellent Student Cadre in Henan Province (province-wide award)	Apr. 2016
Second prize in China Undergraduate Mathematical Modeling Contest	Oct. 2015